

Problem Definition & Data Source Identification

Submission: September 29, 2026 | **Deliverable:** 2–3 page project proposal

Purpose: Establish a well-defined analytics problem and demonstrate that the project has appropriate, accessible data sources before moving into analysis.

Phase 1 Objectives

- **Define the research problem clearly.** State the real-world problem, who is affected, and why the problem is worth analyzing.
- **Identify project goals.** Specify whether the project will focus on prediction, classification, clustering, visualization, or another analytics objective.
- **Identify and evaluate data sources.** Select relevant internal, external, API, sensor, survey, transactional, or other data sources that can support the project.
- **Establish feasibility.** Confirm that the required data can be accessed, understood, and used within the scope and timeline of the term project.

What Your Proposal Should Contain

1. Problem Statement

- Describe the business, organizational, or research problem you intend to investigate.
- Explain the context and why the problem matters.
- Identify the population, process, customers, organization, or system being studied.
- State the problem in a specific and measurable way rather than as a broad topic.

2. Project Objectives & Analytics Goal

- List 2–4 specific objectives for the project.
- Identify the primary analytics task: prediction, classification, clustering, visualization, anomaly detection, or another appropriate approach.
- If applicable, identify the target/outcome you expect to predict or explain.
- Briefly describe what a successful project would produce or enable.

3. Data Source Identification

- List the main data sources you plan to use.
- For each source, explain what information it contains and why it is relevant to the problem.
- Indicate whether each source is internal, external, an API, sensor/device data, survey data, transactional data, public/open data, or another type.
- Identify the expected format and approximate size or time period when known.

4. Data Access, Quality & Feasibility

- Explain how the data will be obtained and whether access is already available.

- Identify obvious issues such as missing values, inconsistent records, limited observations, bias, or data integration challenges.
- Mention any privacy, security, consent, licensing, or ethical considerations that may affect the project.
- Explain why the proposed data sources are sufficient—or what additional data may be needed.

5. Preliminary Project Scope

- Define what the project will and will not attempt to accomplish.
- Identify the expected users or stakeholders of the final analysis.
- Briefly describe the expected final outputs, such as a dashboard, analytical report, predictive model, or decision-support tool.

Data Source Checklist

Source	Type	What it provides	Access / feasibility
Example: Customer database	Internal	Profile, account, transaction information	Available / permission required
Example: Public dataset	External	Relevant demographic or market variables	Public download
Example: Web/API source	API	Current or historical external indicators	API access / usage limits
Example: Device or sensor data	Sensor	Usage, location, performance measures	Access and technical integration

Deliverable & Submission Requirements

- Length: 2–3 pages, excluding references or appendices.
- Include a clear project title and the names of all team members.
- Use headings that correspond to the proposal components above.
- Cite external datasets, APIs, reports, or other sources used to justify the project.
- Use concise, professional writing and clearly distinguish confirmed data access from planned or potential sources.
- Submit one proposal per project/team by the Phase 1 deadline.

Phase 1 Quality Standard

Area	Strong proposal	Needs improvement
Problem	Specific, meaningful, and measurable problem	Broad topic with no clear problem
Analytics goal	Clear task and intended outcome	Unclear or unrealistic objective
Data sources	Relevant, credible, and feasible sources	Sources are vague, inaccessible, or insufficient
Feasibility	Access, quality, ethics, and constraints considered	Major data risks are overlooked

Phase 1 checkpoint: *Before collecting or modeling data, make sure you can clearly explain the problem, the analytics goal, and where the data will come from.*